VLM-DexSafeDagger: VLM-Guided Intervention Policy Towards Safer Visuomotor Skill Distillation

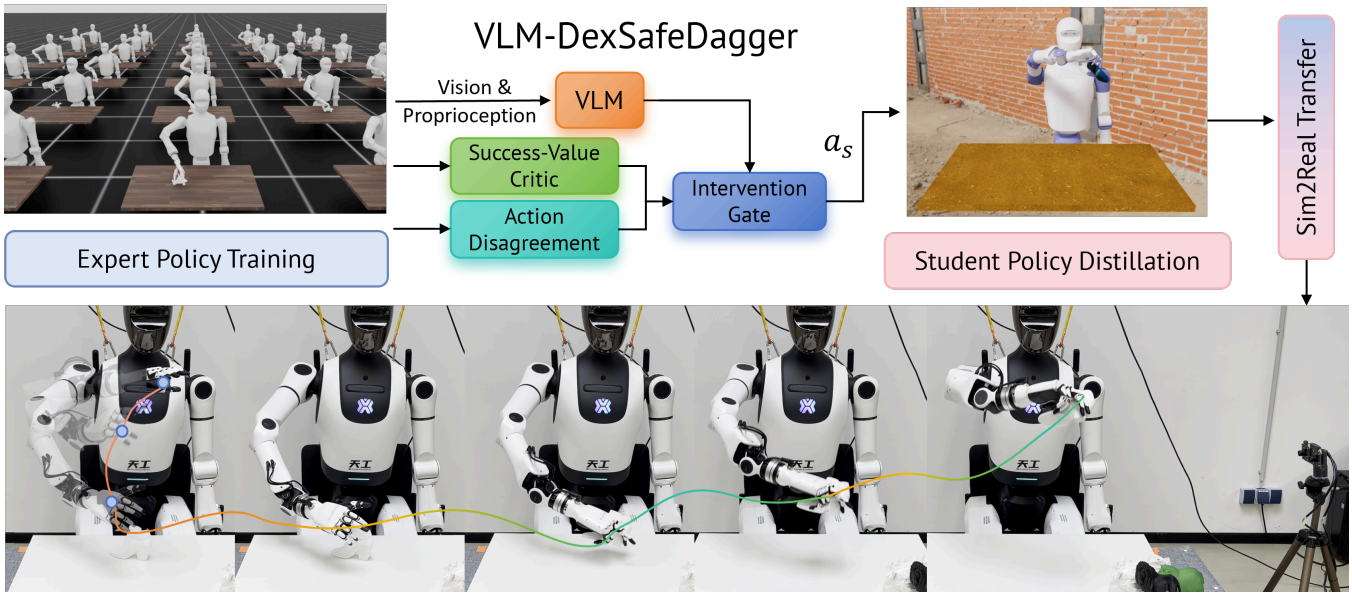


Fig. 1: **VLM-DexSafeDagger overview.** We propose a teacher-intervened online distillation pipeline that condenses teacher skills into a deployable RGB-based student policy. The unsafe exploration rate is substantially suppressed in the early training phase via a proposed VLM-guided intervention policy that combines teacher–student action disagreement with a Bellman success-value critic trained on safe-lift-success labels, while a vision-language model maintains an evolving, safety-aware understanding of the scene and periodically adjusts the intervention policy during online distillation.

***Abstract*— Learning dexterous manipulation is challenging due to high-dimensional action spaces and complex contact dynamics. In addition, safety-unaware exploration during learning can lead to hardware damage and unstable or even hazardous contact events, posing a significant barrier to real-world deployment. To address this challenge, we propose VLM-DexSafeDagger, a safety-aware online skill-distillation framework that transfers reinforcement learning (RL) teacher policies to a vision-based student policy. VLM-DexSafeDagger combines an ℓ_2 teacher–student disagreement gate with a learned Bellman success-value critic that estimates the prospective safe-lift-success value of the current student state–action pair. A vision-language model periodically adjusts the intervention policy parameterized by disagreement limit and acceptable predicted success rate. This design enables safer exploration during online interactive imitation learning using high-level visual context. Experiments on grasp-and-lift tasks show that VLM-DexSafeDagger significantly improves training safety, reducing unsafe episodes by up to 80% during the training period compared with vanilla DAgger-based distillation. Meanwhile, the distilled student policy achieves comparable lift success rates and demonstrates successful transfer from simulation to the real robot.**

I. INTRODUCTION

Dexterous robotic hands typically possess a large number of degrees of freedom (DoFs) and exhibit complex contact dynamics, enabling them to perform fine-grained and diverse manipulation behaviors [1]–[4]. In recent years, reinforcement learning (RL)-based approaches [3], [5]–[7] have made significant progress in addressing the complex control problems posed by high-dimensional action spaces and intricate contact dynamics in dexterous manipulation tasks.

Despite these advances, existing RL frameworks primarily optimize for task success rates and rarely incorporate explicit modeling or constraints related to safety. During training, common failure modes, including severe collisions between the hand and the environment, unintended object ejection due to excessive contact forces, and loss of stable hand posture [8], [9], not only reduce training efficiency but also introduce potential risks during real-world deployment. Therefore, how to reduce unsafe exploration while maintaining learning efficiency remains a critical challenge in dexterous manipulation.

To bridge the sim-to-real gap, skill distillation is widely used to transfer the behavior of expert policies trained with privileged information to student policies capable of operating on real robots. However, existing distillation methods [10], [11] typically rely on behavior cloning or the standard DAgger framework for data aggregation. Their primary objective is to mitigate performance degradation caused by distribution

shift, rather than explicitly enforcing safety constraints during student policy learning. During distillation, the student policy may deviate from the teacher's state–action distribution and enter regions that are either insufficiently covered by the teacher or inherently high-risk, leading to unsafe behaviors such as severe collisions or unstable hand configurations. Therefore, introducing explicit safety mechanisms into the distillation process—so that high-risk exploration can be avoided without sacrificing learning efficiency—remains an important and largely underexplored problem. Typical safety-aware distillation approaches, such as SafeDAgger [12], have primarily focused on autonomous driving scenarios and have not been systematically studied in contact-rich tasks such as dexterous manipulation.

The contributions of this work are summarized as follows:

- We propose VLM-DexSafeDagger, a VLM-guided, safety-sensitive framework for distilling RL-acquired teacher policies into a single RGB-based student policy. To the best of our knowledge, this is the first attempt in dexterous manipulation to incorporate a mixed intervention policy combining teacher–student action disagreement with a Bellman success-value critic, while the VLM reasons about scene-level safety from rollout statistics and representative visual observations to regulate the intervention gate.
- We empirically demonstrate that VLM-DexSafeDagger, which combines disagreement, success-value prediction, and VLM-guided intervention policy adaptation, substantially reduces unsafe exploration compared with Vanilla DAgger, achieving an approximately **80%** relative reduction in the unsafe-episode rate during early training. It also reduces the average unsafe-episode rate by **7.5 percentage points** relative to disagreement-only SafeDAgger [12]. Ablation results further demonstrate the effectiveness of VLM-guided adaptation across different visual contexts and unsafe scenarios: VLM-DexSafeDagger consistently achieves lower unsafe-episode rates than DexSafeDagger, with an average reduction of **4 percentage points**.

II. Related work

Skill Distillation. Skill distillation transfers behavior from a high-capacity teacher to a compact student and has been widely adopted in both computer vision and robotics. In robotics, the teacher can leverage privileged state and simulator access during training, while the student is distilled to operate from on-board observations (e.g., RGB). DexTrAH-RGB [11] distills a privileged RL teacher into an RGB-based policy via DAgger-based imitation learning, while UniGraspTransformer [13] distills offline trajectories from object-specific teachers into a unified Transformer policy.

Adaptive Intervention in Interactive Imitation Learning. Many imitation learning methods use fixed offline datasets for supervision [14], [15]. In sequential decision-making, however, distribution shift induces covariate shift and compounding errors. Interactive imitation learning [16] mitigates these issues by collecting online supervision along trajectories visited by the learner. DAgger [10] addresses covariate shift by iteratively relabeling learner-visited states with an expert, while SafeDAgger [12] adds teacher intervention in high-risk states to reduce unsafe exploration.

Later methods introduce adaptive intervention mechanisms based on uncertainty-driven rules [17], human gating [18], or learned intervention criteria [19]. However, they have primarily been evaluated in relatively simple control or autonomous-driving settings. Fully autonomous expert relabeling during online distillation remains rare in robotic manipulation, especially for dexterous, contact-rich tasks.

Vision-Language-Guided Robot Learning. Visual and semantic information is widely recognized as a strong cue for embodied agents to interpret task specifications and reason about unstructured environments [20], [21]. One line of work uses VLMs to generate demonstrations or supervisory signals for offline policy construction, such as Manipulate-Anything [22]. Complementary approaches, including RACER [23] and FOREWARN [24], employ VLMs to guide risk prediction, provide recovery guidance for unsafe situations, or govern action selection during online policy learning or real-time execution.

III. Problem Statement and Methods

A. Problem Formulation

We consider online interactive imitation learning with teacher intervention in a partially observable Markov decision process (POMDP) $\mathcal{M} = (\mathcal{S}, \mathcal{A}, \mathcal{O}, P, \Omega, r, \gamma)$, with privileged teacher state $s_t \in \mathcal{S}$, deployable student observation $o_t \in \mathcal{O}$ (RGB), action $a_t \in \mathcal{A}$, observation model $o_t \sim \Omega(\cdot \mid s_t)$, and dynamics $s_{t+1} \sim P(\cdot \mid s_t, a_t)$. During training, we assume access to a set of privileged teacher policies $\{\pi_{T_i}\}_{i=1}^{K}$, whereas at deployment the student policy π_S must act solely from o_t. Our goal is to distill the collective teacher behavior into an RGB-based student that achieves high task success while minimizing unsafe exploration during training.

The induced rollout distribution $p_{\pi_\mathcal{G}}$ is defined by the gated rollout policy $\pi_\mathcal{G}$, which switches between the teacher and student policies according to the intervention gate. Its learning objective and intervention gate are

$$\begin{aligned}
\min_{\substack{\pi_S \in \Pi_S \\ \mathcal{G} \in \mathfrak{G}}} \Bigg[& \underbrace{-\mathbb{E}_{\tau \sim p_{\pi_\mathcal{G}}}[\mathbb{I}_{\mathrm{succ}}(\tau)]}_{\text{maximize success}} + \lambda \underbrace{\mathbb{E}_{\tau \sim p_{\pi_\mathcal{G}}}\left[\sum_{t=0}^{T-1} \mathbb{I}_{\mathrm{unsafe}}(s_t, a_t)\right]}_{\text{minimize unsafe}} \Bigg] \\
\text{s.t.} \quad & a_t = g_t a_t^T + (1 - g_t) a_t^S, \\
& g_t = \mathcal{G}(s_t, o_t; \boldsymbol{\tau}^{(k)}) \in \{0, 1\}, \\
& s_{t+1} \sim P(\cdot \mid s_t, a_t), \qquad o_t \sim \Omega(\cdot \mid s_t).
\end{aligned} \tag{1}$$

where $g_t = 1$ indicates teacher takeover and $g_t = 0$ denotes student control. The intervention policy is parameterized by $\boldsymbol{\tau}^{(k)} = (\tau_\delta^{(k)}, \tau_Q^{(k)})$, which governs teacher takeover decisions in $\mathcal{G}$. Periodically, a VLM proposes a displacement $\Delta\boldsymbol{\tau}^{(k)}$, from which the policy is updated to adjust to distributional shift.

We formalize task success via a trajectory-level predicate $\mathbb{I}_{\mathrm{succ}}(\tau) \in \{0, 1\}$, which is 1 if the object is grasped, lifted, and held above a height threshold for a consecutive duration.

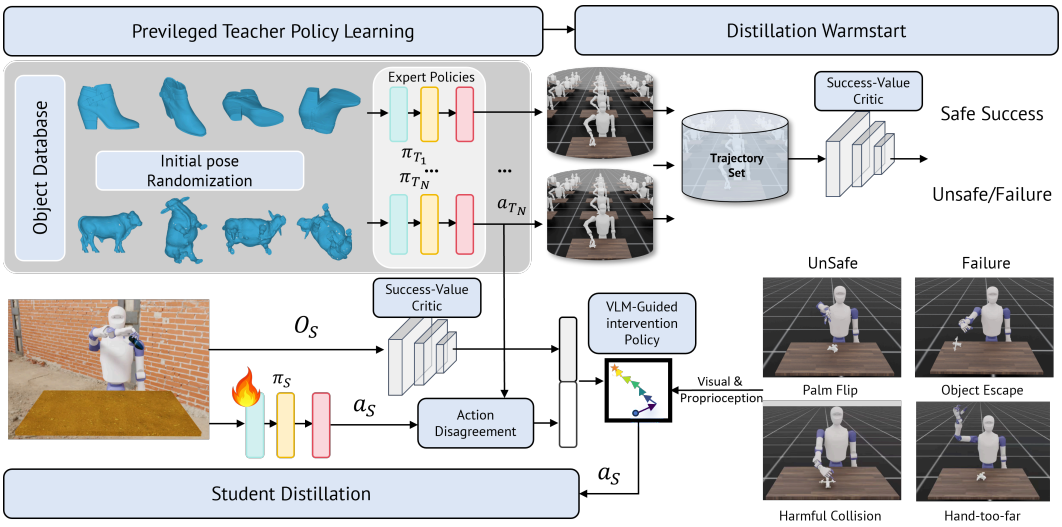


Fig. 2: Overview of the VLM-DexSafeDagger teacher–student learning pipeline. A privileged reinforcement learning teacher is trained in simulation and subsequently distilled into an RGB-based visuomotor policy for real-world deployment. A vision-language model periodically proposes a displacement of the intervention policy's operating point; deterministic clipping and smoothing constrain the resulting policy update.

Safety is governed by early termination conditions: letting $\mathbb{I}_{\text{unsafe}}(s_t, a_t) \in \{0, 1\}$ indicate whether any unsafe event (e.g., harmful collision, object escape, palm flip, hand-too-far) occurs at time t, the trajectory is safe if $\prod_{t=0}^{T-1} \big(1 - \mathbb{I}_{\text{unsafe}}(s_t)\big) = 1$.

B. Overview of the Distillation Pipeline

Fig. 2 summarizes VLM-DexSafeDagger, our teacher–student distillation pipeline, which consists of five core components: (i) privileged teacher policy training in simulation (Sec. C), (ii) Bellman success-value critic learning to estimate safe lift success under student actions (Sec. D), (iii) deterministic teacher intervention (Sec. E), (iv) VLM-guided intervention policy adaptation (Sec. F), and (v) online student skill distillation (Sec. G).

The privileged teacher is first trained in simulation, and its rollouts provide an offline warm start for the Bellman critic. The critic estimates prospective lift success under the student action and continues to be updated from a ring replay buffer.

During online distillation, the teacher intervenes when either teacher–student action disagreement is high or the critic predicts low success. The student is then optimized online from teacher labels collected under the resulting rollout policy $\pi_{\mathcal{G}}$. A VLM is periodically queried to propose updates to the intervention policy's operating point. The resulting candidate operating point is clipped and smoothed before use. Pseudo-code is provided in Alg. 1.

Algorithm 1 VLM-DexSafeDagger

Train the privileged teacher policy π_T via RL
Collect a warm-start dataset $\mathcal{D}_0$ under teacher control
Pretrain the Bellman critic $\hat{Q}_{\text{succ}}$ on $\mathcal{D}_0$
Initialize the intervention policy parameters $\boldsymbol{\tau}^{(0)} = (\tau_\delta^{(0)}, \tau_Q^{(0)})$
Initialize the visual buffer $\mathcal{B}_V$
for iteration $k = 1 \dots N$ **do**
  Observe the teacher state s_t and student observation o_t
  Compute the actions $a_t^T \sim \pi_T(\cdot|s_t)$ and $a_t^S \sim \pi_S(\cdot|o_t)$
  Compute the action disagreement $\delta_t = \|a_t^S - a_t^T\|_2$
  Predict the safe-lift-success value $\hat{Q}_{\text{succ}}(s_t, a_t^S)$
  Evaluate the intervention gate
    $g_t \leftarrow \mathbb{I}[\delta_t > \tau_\delta \ \vee \ \hat{Q}_{\text{succ}}(s_t, a_t^S) < \tau_Q]$
  Execute the gated action $a_t = g_t a_t^T + (1 - g_t) a_t^S$
  Store the transition in the critic replay buffer $\mathcal{D}$
  Update $\hat{Q}_{\text{succ}}$; if $g_t = 1$, update π_S
  Add representative visual states to $\mathcal{B}_V$
  if an intervention policy update is due **then**
    Summarize rollout context $\mathcal{C}_k$ and sample images $\mathcal{V}_k$
    Query the VLM for the shift $\Delta\boldsymbol{\tau}^{(k)}$ given $(\mathcal{C}_k, \mathcal{V}_k)$
    Form the candidate $\tilde{\boldsymbol{\tau}}^{(k)} = \boldsymbol{\tau}^{(k)} + \Delta\boldsymbol{\tau}^{(k)}$
    Clip and smooth $\tilde{\boldsymbol{\tau}}^{(k)}$ to obtain $\boldsymbol{\tau}^{(k+1)}$
  end if
end for

C. Privileged Teacher Policy Training

Teacher policy learning is formulated as a reinforcement learning problem with transition tuples $\langle s_t, a_t, r_t, s_{t+1} \rangle$, where s_t is the privileged state, a_t is the action, r_t is the reward, and s_{t+1} is the next state. The objective is to maximize the expected discounted return,

$$\pi_T^* = \arg\max_{\pi_T} \mathbb{E}_{\pi_T} \left[\sum_{t=0}^{T-1} \gamma^t r_t \right]. \tag{2}$$

We use a shaped reward $r_t = \sum_{k=1}^{K_r} r_{t,k}$ to encourage stable grasp acquisition, object lifting, and goal reaching (Table I). Our teacher-training control interface follows the normalized direct motor-command formulation used in UniDexGrasp++ [25]. Unlike the human-motion-derived hand-action manifold and FABRICS interface used by DexTrAH-

TABLE I: Reward components used for teacher policy training.

Term	Name	Purpose
r_{dist}	Hand distance	Guides the hand toward the object.
r_{curl}	Finger curl	Encourages a grasp-ready finger posture.
r_{align}	Palm alignment	Aligns the palm to a palm-down pose.
r_{contact}	Contact bonus	Rewards stable thumb–finger contact.
r_{near}	Stay-close	Keeps the hand near the object after reaching.
r_{goal}	Goal tracking	Guides the grasped object toward the goal.
r_{lift}	Lift	Rewards lifting after stable contact.
r_{action}	Action smoothness	Penalizes abrupt action changes.

RGB [11], our 13 normalized outputs map directly to joint-position targets for seven arm and six hand DoFs.

The teacher is trained in simulation using an asymmetric actor–critic setup. Details of the actor and critic observation spaces are provided in the Appendix. The actor receives noisy and partial proprioceptive observations that approximate realistic sensing, while the critic is provided with the privileged full state of both the robot and object, available only in simulation.

Episodes terminate at

$$T = \min\{t \,|\, \mathbb{I}_{\text{unsafe}}(s_t) = 1 \;\text{ or }\; t \geq T_{\max}\}, \tag{3}$$

where $T_{\max} = 10\,\text{s}$ is the timeout horizon. Early termination conditions include palm flip, harmful collision, object escape, and hand-too-far (see Sec. A), and are treated as unsafe events for both teacher training and subsequent intervention decisions.

D. Bellman Success-Value Critic

ℓ_2 action disagreement alone is insufficient as an intervention signal and may miss small action deviations that could cause unstable grasps, collisions, or object escape. We therefore learn a Bellman critic, trained with safe-lift-success labels, that estimates the probability of safe lift success for the current student state–action pair and provides a prospective signal complementary to action disagreement.

The Bellman critic evaluates a multimodal state–action pair (s_t, a_t^S), where s_t combines robot proprioception and embeddings from a visual encoder. The encoder is trained end-to-end during skill distillation and regularized by the auxiliary object-position prediction loss in Sec. G. The twin critics Q_1 and Q_2 are ReLU MLPs with hidden dimensions $[256, 128]$. Each critic's logit $z_i(s,a)$ is mapped to a success probability by $Q_i(s,a) = \sigma(z_i(s,a)/T)$, with calibration temperature $T = 4$. We use the conservative estimate $\hat{Q}_{\text{succ}}(s,a) = \min_{i\in\{1,2\}} Q_i(s,a)$; lower values indicate higher prospective risk.

The critics are pretrained on teacher transitions and updated online from a ring replay buffer. The label $u_t \in \{0,1\}$ indicates safe lift success: it is 1 only when the lift criterion is satisfied without an unsafe event. Training uses the SARSA-style target

$$y_t = \text{clip}\Big(u_t + \gamma(1-d_t^{\text{eff}}) \min_{i\in\{1,2\}} Q_i^{\text{targ}}(s_{t+1}, a_{t+1}),\, 0,\, 1\Big), \tag{4}$$

where d_t is the termination flag, h_t indicates whether a_{t+1} is valid, $d_t^{\text{eff}} = \max(d_t, 1-h_t)$, and Q_i^{targ} are slowly updated target networks. Both critics minimize the mean-squared error (MSE) between their predictions and y_t; the minimum across critics reduces overestimation.

E. Intervention Policy

The intervention policy uses a deterministic two-channel gate. The first channel measures teacher–student disagreement, $\delta_t := \|a_t^S - a_t^T\|_2$, where $a_t^T \sim \pi_T(\cdot \mid s_t)$ and $a_t^S \sim \pi_S(\cdot \mid o_t)$. The second channel evaluates prospective task success through $\hat{Q}_{\text{succ}}(s_t, a_t^S)$. The intervention policy is parameterized by $\boldsymbol{\tau}^{(k)} = (\tau_\delta^{(k)}, \tau_Q^{(k)})$, where $\tau_\delta^{(k)}$ is the maximum admissible disagreement and $\tau_Q^{(k)}$ is the minimum acceptable predicted success.

These parameters define the deterministic intervention gate:

$$g_t = \mathbb{I}\Big[\hat{Q}_{\text{succ}}(s_t, a_t^S) < \tau_Q^{(k)} \;\vee\; \delta_t > \tau_\delta^{(k)}\Big]. \tag{5}$$

Thus, either large disagreement or low predicted success triggers teacher takeover. When the two signals conflict, the conservative OR rule applies: low teacher–student disagreement does not suppress intervention if the critic predicts low student success, and a favorable critic prediction does not suppress intervention if the action disagreement is large. The student acts only when neither intervention condition is triggered. Teacher takeover is treated as empirically safer rather than as an absolute safety guarantee, and the resulting safety performance remains bounded by the teacher policy's own performance ceiling. Decreasing τ_δ or increasing τ_Q makes the intervention policy more conservative; the opposite shift increases student autonomy.

We empirically initialize $\tau_\delta^{(0)}$ and $\tau_Q^{(0)}$ from the 90th percentile of warm-start disagreement and the 10th percentile of predicted success over successful teacher rollouts, respectively. These values define the initial operating point $\boldsymbol{\tau}^{(0)}$.

F. VLM-Guided Intervention Policy Adaptation

During online distillation, the student policy, visited-state distribution, and Bellman critic's calibration co-evolve, and the dominant failure modes may also change. A stationary intervention policy therefore cannot maintain a consistent safety–autonomy trade-off. We periodically query a VLM to guide the displacement of the intervention policy's operating point. At iteration k, the VLM receives rollout statistics $\mathcal{C}_k$, including disagreement (δ_t), predicted success ($\hat{Q}_{\text{succ}}(s_t, a_t^S)$), intervention rate ($\beta$), and unsafe-episode rate (UnsafeRateAvg). It also receives a representative image batch $\mathcal{V}_k$ sampled from the visual buffer $\mathcal{B}_V$ to cover high-disagreement, low-success, and unsafe states. From this context, it proposes an operating-point displacement $\Delta\boldsymbol{\tau}^{(k)} = (\Delta\tau_\delta^{(k)}, \Delta\tau_Q^{(k)})$, producing the candidate operating point $\tilde{\boldsymbol{\tau}}^{(k)} = \boldsymbol{\tau}^{(k)} + \Delta\boldsymbol{\tau}^{(k)}$.

Each candidate policy parameter is clipped around its initial value and exponentially smoothed:

$$\tau_x^{(k+1)} = (1-\alpha)\tau_x^{(k)} + \alpha\ \text{clip}\Big(\tilde{\tau}_x^{(k)}, c_x^{\min}\tau_x^{(0)}, c_x^{\max}\tau_x^{(0)}\Big), \quad x \in \{\delta, Q\}. \tag{6}$$

Here, α controls the adaptation rate, and $c_x^{\min}$ and $c_x^{\max}$ define the admissible calibration region around $\tau_x^{(0)}$.

G. Student Skill Distillation

Student learning is posed as supervised imitation from teacher-labeled transitions collected under the gated rollout policy $\pi_{\mathcal{G}}$. The student observes RGB images (two stereo views) together with robot proprioception.

We optimize a combined objective $\mathcal{L} = \mathcal{L}_{\text{action}} + \mathcal{L}_{\text{aux}}$, where the auxiliary loss predicts object position to regularize the visual encoder, $\mathcal{L}_{\text{aux}} = \|\hat{\mathbf{x}}_{\text{obj}} - \mathbf{x}_{\text{obj}}\|_2$.

For action imitation, we use a weighted L_2 loss, $\mathcal{L}_{\text{action}} = \|\mathbf{a}_S - \mathbf{a}_T\|_W^2 = (\mathbf{a}_S - \mathbf{a}_T)^\top W(\mathbf{a}_S - \mathbf{a}_T)$, with $W = \text{diag}(w_1, \ldots, w_d)$. We set $w_j = \sigma_{0,j}^{-2}$ using a reference scale $\sigma_{0,j}$ (e.g., the teacher action standard deviation on dimension j), yielding $\mathcal{L}_{\text{action}} = \sum_{j=1}^{d}(a_{S,j} - a_{T,j})^2/\sigma_{0,j}^2$.

Training proceeds in two stages. (i) *Warm-start:* we collect an offline dataset (e.g., 100k transitions) under full teacher control ($g_t \equiv 1$) to pretrain the Bellman critic. (ii) *Online distillation:* we perform interactive rollouts under $\pi_{\mathcal{G}}$ and query teacher labels on intervened steps. The student is updated online using the current-step teacher label only; no replay buffer or offline dataset is maintained for student updating. The design reduces the effect of stale supervision and stabilizes online imitation updates.

IV. Experiment

A. Evaluation Metrics

We designed the following metrics to evaluate both teacher and student policy performance. Let $\tau \in \{1, \ldots, N\}$ index evaluation episodes and $t \in \{0, \ldots, T_\tau\}$ index timesteps. Unless otherwise stated, we run the evaluation episodes with a horizon of 10 s.

a) Lift success (ever-lifted with hold).

Let h_t^τ be the object height above the table and H a lift threshold. An episode is considered successful if the object remains above H for at least N_{hold} consecutive steps. We define $\mathbb{I}_{\text{lift}}^\tau = \mathbb{I}\big[\exists\, t:\ \forall k \in \{0, \ldots, N_{\text{hold}} - 1\},\ h_{t+k}^\tau \geq H\big]$ and report: $\text{LiftSucc} = \frac{1}{N}\sum_{\tau=1}^{N} \mathbb{I}_{\text{lift}}^\tau$.

b) Unsafe episode rate.

As stated above, an episode may terminate early due to one of the following conditions: palm flipped ($\mathbb{I}_{\text{flip}}^\tau$), harmful collision ($\mathbb{I}_{\text{coll}}^\tau$), object escape ($\mathbb{I}_{\text{obj}}^\tau$), or hand too far ($\mathbb{I}_{\text{hf}}^\tau$). Let $\mathcal{M} = \{\text{coll}, \text{obj}, \text{flip}, \text{hf}\}$ denote the set of termination modes, and let $\mathbb{I}_m^\tau \in \{0, 1\}$ indicate whether mode $m \in \mathcal{M}$ occurs in episode τ. An episode is unsafe if any termination mode is triggered: $\mathbb{I}_{\text{unsafe}}^\tau = \mathbb{I}\big[\bigvee_{m\in\mathcal{M}} \mathbb{I}_m^\tau = 1\big]$. We report: $\text{UnsafeRateAvg} = \frac{1}{N}\sum_{\tau=1}^{N} \mathbb{I}_{\text{unsafe}}^\tau$.

c) Failure-mode decomposition.

Each episode is assigned a single terminal mode $m^\tau \in \mathcal{M}$ from the set of unsafe events (collision, object escape, palm flip, hand-too-far) as well as timeout/success where applicable. We define $\mathbb{I}_m^\tau = \mathbb{I}[m^\tau = m]$ and report $\text{FailureModeRate}(m) = \frac{1}{N}\sum_{\tau=1}^{N} \mathbb{I}_m^\tau, m \in \mathcal{M}$.

When multiple unsafe conditions are triggered in the same timestep, we record the first triggered termination condition (using a fixed priority order) as the episode's failure mode.

TABLE II: Teacher policy performance on four illustrative objects from the 40-object training set (200 evaluation episodes per object). Mean is computed over all 40 objects.

Metric (%)	WindCart	Boot	ToolBox	Cow	**Mean**
Lift Success	93.5	100.0	88.5	87.0	91.7
Unsafe Rate Avg	6.5	13.5	13.5	13.5	11.4
Harmful collision	0.0	0.0	6.5	0.0	2.1
Object escape	6.5	6.5	6.5	13.5	8.0
Palm flipped	0.0	6.5	0.0	0.0	1.2
Hand too far	0.0	0.0	0.0	0.0	0.1

B. Simulation Experiment Setup

Simulation uses single-arm control of a Tiangong 2 Pro humanoid with an InspireHand RH56DfX and a 13-D action space (7 arm + 6 hand DoF). The RGB student receives two 320×240, 30-fps camera streams, encoded by a ResNet-18 and fused into a 128-D feature appended to the actor observation. Teacher training and student distillation run in Isaac Sim 5.0.0 / Isaac Lab 2.2.1 on one RTX 5090 GPU.

We train 40 object-specific teachers on Omniverse Asset Library objects, each for 12,000 episodes (3.5 h), totaling 480,000 episodes and approximately 140 GPU-hours. Lift-success rates span 72–100% and unsafe rates 5–30%; Table II shows four illustrative objects. The joint student is distilled across all 40 objects for 100,000 episodes (2.5 h), using DexTrAH-RGB-style randomization of lighting intensity and direction, background textures, object materials, and table appearance.

We use GPT-5.5 with high reasoning effort as the VLM. We issue the first VLM query after 2,000 episodes and subsequently query it every 10,000 distillation episodes, resulting in 10 VLM calls over the full 100,000-episode training run and approximately 70,000 tokens of total usage per run.

C. Ablation Studies

We perform the following ablation studies to validate the design choices in VLM-DexSafeDagger and answer three questions: **(i) Does the intervention mechanism in DexSafeDagger enable safer exploration during policy distillation? (ii) Does augmenting action disagreement with success-value prediction improve the intervention signal? (iii) Is vision-language reasoning necessary for adapting the intervention boundary?**

All retained ablations use the same teacher-training setup from Sec. III-C and differ only in the student-distillation and intervention mechanisms.

(a) Vanilla DAgger. The student is trained using teacher supervision, with no teacher intervention during rollouts ($\beta = 0$).

(b) Vanilla SafeDAgger. We replace Vanilla DAgger with a disagreement-gated SafeDAgger baseline. The teacher intervenes only when $\|a_t^T - a_t^S\|_2$ exceeds a fixed threshold.

(c) DexSafeDagger. This setting extends (b) with a Bellman success-value critic under a stationary intervention policy. Teacher intervention is triggered by either large teacher–student disagreement or low predicted success.

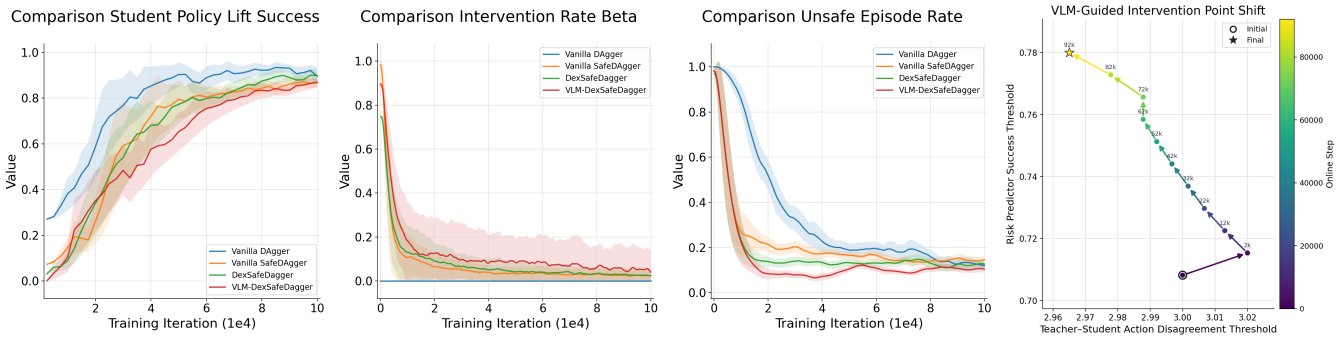


Fig. 3: **Ablation studies.** The first three panels compare four distillation variants in terms of student lift success, teacher intervention rate β, and unsafe-episode rate, respectively, as they evolve during online distillation. The rightmost panel shows the VLM-guided trajectory of the intervention operating point. Shaded regions indicate variability across runs (mean $\pm$ std). Curves correspond to vanilla DAgger (blue), vanilla SafeDAgger (orange), DexSafeDagger (green), and VLM-DexSafeDagger (red).

TABLE III: Performance of ablations (a)–(f) on four illustrative objects from the 40-object training set (200 post-training evaluation episodes per object). Each ablation reports lift success and average unsafe rate. Mean is computed over all 40 objects.

Metric (%)	WindCart		Boot		ToolBox		Cow		**Mean**	
	Lift	Unsafe	Lift	Unsafe	Lift	Unsafe	Lift	Unsafe	Lift	Unsafe
(a)	88.5	13.5	89.0	20.5	82.5	19.0	88.0	17.5	87.1	17.4
(b)	88.0	12.0	89.5	19.5	81.5	18.5	87.0	16.0	86.3	16.7
(c)	88.0	7.5	90.0	15.0	82.0	15.5	88.0	12.0	86.8	11.9
(d)	88.5	6.0	90.0	10.5	82.5	11.0	88.5	8.5	87.5	9.3
(e)	88.0	7.5	89.5	12.5	82.0	13.0	88.0	10.5	86.9	10.2
(f)	86.5	7.0	88.5	14.0	81.0	13.5	87.0	11.0	85.7	11.7

(d) VLM-DexSafeDagger. This setting extends (c) by replacing the stationary intervention policy with the proposed VLM tuner, which periodically calibrates the disagreement-risk intervention policy using rollout statistics and representative visual states.

(e) LLM-DexSafeDagger. This setting matches (d), except that the language model receives only the rollout statistics and no visual observations.

(f) DexSafeDagger + Adaptive Intervention under a Performance Curriculum. We replace the VLM in (d) with an adaptive threshold curriculum inspired by ThriftyDAgger [26] and driven by recent lift-success and unsafe-episode rates, dynamically balancing conservative teacher intervention against aggressive student exploration throughout training. All other components and the threshold-update schedule are held fixed.

We first address question (i) by examining the distillation behavior in Fig. 3. Ablations (a)–(d) achieve comparable final lift success (left), consistent with the intervention policy's focus on safety rather than task success. Unlike Vanilla DAgger ($\beta = 0$), the three intervention variants reduce teacher intervention as the student improves (middle). They reduce early unsafe episodes by approximately 80% relative to Vanilla DAgger and remain below 20% throughout distillation, with VLM-DexSafeDagger achieving the lowest rate overall (right).

We address question (ii) by comparing ablation (b), which uses disagreement alone, with ablation (c), which adds success-value prediction while retaining the same RL teacher and stationary intervention policy. The combined gate intervenes in low-predicted-success states missed by disagreement alone, reducing the average unsafe-episode rate by about 7.5 percentage points and limiting collision and object-escape failures without sacrificing task performance. Its unsafe rate approaches the teacher's average ($\approx 12\%$) by the end of training and reaches a minimum of approximately 8%. Fig. 4 further shows that harmful collisions and object escape dominate unsafe episodes. Compared with Vanilla DAgger, all three intervention variants suppress both failure modes, particularly their early spikes, demonstrating that teacher intervention limits unsafe exploration during the initial distillation phase.

Finally, we address question (iii) by comparing ablations (c), (f), (e), and (d), which form a progression from stationary intervention, to performance-curriculum adaptation, to semantics-guided LLM adaptation, and finally to our VLM-guided recipe. This controlled ladder isolates the contributions of feedback-based threshold adaptation, semantic reasoning, and visual context. As shown in Table III, VLM-guided adaptive intervention consistently achieves the highest lift success and the lowest unsafe rate across the evaluated objects. Both LLM-guided and curriculum-based adaptation improve overall safety, but neither matches the full VLM-guided recipe. The curriculum-based intervention also incurs a small reduction in lift success.

D. Real-World Experimental Results

We evaluate the deployed student policy on 22 objects, comprising 14 seen objects and 8 unseen objects. Table IV reports a representative six-object subset alongside pooled results over the complete sets of 14 seen and 8 unseen objects. The real-world results show lower lift-success rates and higher unsafe-episode rates than the simulation results, indicating that VLM-DexSafeDagger does not fully eliminate the sim-to-real gap. Nevertheless, the physical experiments demonstrate

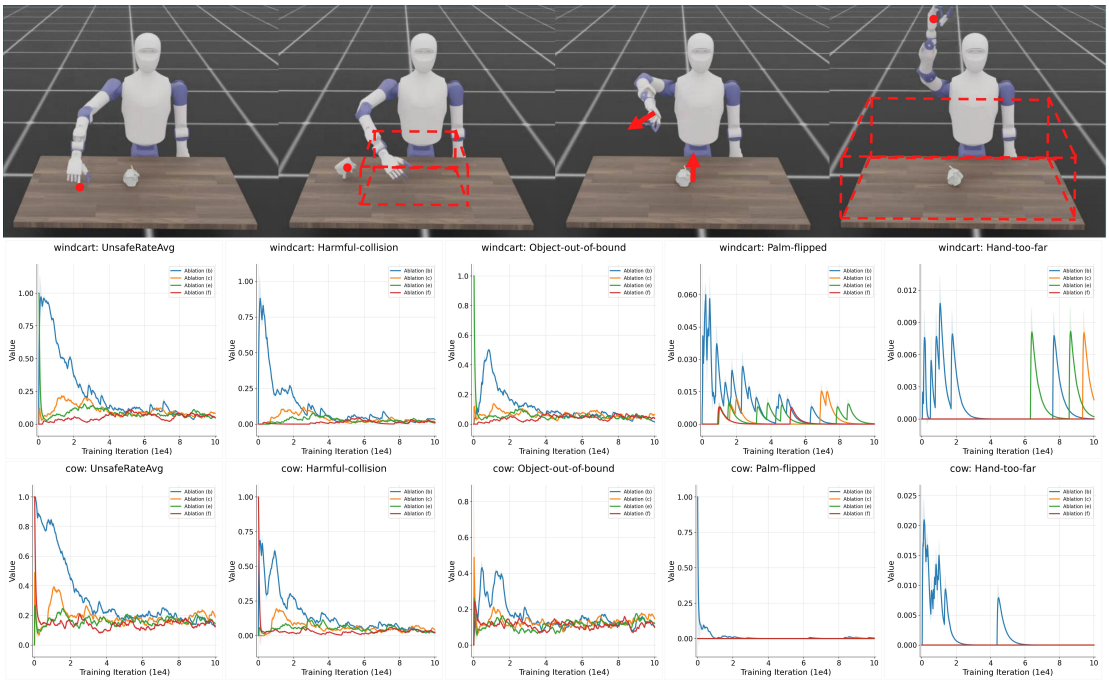


Fig. 4: **Per-object failure-mode decomposition during online distillation ablations.** We illustrate four task-specific failure modes in simulation (harmful collision, object escape, palm flipped, and hand too far) and report the aggregate unsafe rate together with its per-mode breakdown over training iterations. Results are shown for two objects (`windcart` and `cow`); lower is better in all plots. Curves correspond to four distillation variants: blue, Vanilla DAgger [10]; orange, Vanilla SafeDAgger [12]; green, DexSafeDagger; and red, VLM-DexSafeDagger.

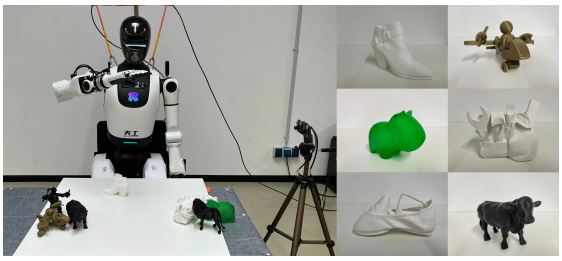


Fig. 5: **Real-world experimental setup and test objects.** Left: the single-arm manipulation platform used for deployment and evaluation. Right: examples of the physical objects used in the real-world lift experiments.

that the policy can transfer successfully from simulation to the real robot.

Unseen objects show a clear generalization gap: pooled lift success drops from 83.6% to 64.6%, while the unsafe rate rises from 14.7% to 39.2%, mainly due to more harmful collisions (4.7% to 18.8%) and object escapes (8.1% to 15.8%). Among the six representative objects shown, no palm-flip or hand-too-far events are observed, suggesting that their real-world failures are dominated by contact-rich interactions rather than gross pose instabilities. Overall, the results indicate that the remaining sim-to-real gap on unseen geometries is largely attributable to a contact-strategy mismatch, i.e., the policy fails to adapt its contact interactions to novel object shapes.

TABLE IV: **(a):** Representative six-object subset of the 22-object real-world evaluation (30 episodes per object across seeds 40–44), with pooled results over the complete sets of 14 seen and 8 unseen objects in the last two columns, and **unseen objects** are shaded; **(b):** real-world performance comparison of ablations (c) and (d).

(a)

Metric (%)	WindCart	Boot	ToolBox	Cow	**Plane**	**Shoe**	**Seen avg. (14)**	**Unseen avg. (8)**
Lift Success	90.0	83.3	80.0	86.7	**66.7**	**60.0**	83.6	**64.6**
Unsafe Rate Avg	13.3	16.7	20.0	23.3	**40.0**	**40.0**	14.7	**39.2**
Harmful collision	0.0	3.3	13.3	13.3	**20.0**	**26.7**	4.7	**18.8**
Object escape	13.3	13.3	6.7	10.0	**20.0**	**13.3**	8.1	**15.8**
Palm flipped	0.0	0.0	0.0	0.0	**0.0**	**0.0**	1.4	**3.3**
Hand too far	0.0	0.0	0.0	0.0	**0.0**	**0.0**	0.5	**1.3**

(b)

Metric (%)	Ablation (c)		Ablation (d)	
	Seen (14)	Unseen (8)	Seen (14)	Unseen (8)
Lift success	83.4	61.7	83.6	64.6
Unsafe-event rate	16.1	41.4	14.8	39.2

The real-world comparison between ablations (c) and (d) further shows that our VLM-guided method lowers the unsafe-event rate on both seen (16.1% to 14.8%) and unseen objects (41.4% to 39.2%) while maintaining a comparable task success rate.

V. Conclusion and Future Work

In this work, we propose VLM-DexSafeDagger, which integrates teacher–student disagreement with a Bellman success-value critic trained on safe-lift-success labels and uses a VLM to adapt the intervention policy. VLM-DexSafeDagger improves exploration safety during multi-teacher distillation, reducing the unsafe-episode rate by **80%** relative to the vanilla DAgger baseline during early training. The introduction of the Bellman critic reduces the average unsafe-episode rate by **7.5 percentage points** relative to the disagreement-only SafeDAgger baseline. In addition, VLM-guided intervention policy adaptation yields consistent improvements across different visual contexts and unsafe scenarios, reducing the average unsafe-episode rate by a further **4 percentage points** relative to DexSafeDagger. The proposed method approaches and, in some cases, surpasses the teacher policy's safety performance. Successful real-world deployment shows that the resulting student policy partially bridges the sim-to-real gap.

A clear limitation is that performance during distillation, as well as the final student policy, is largely constrained by teacher capability. Developing a safety-aware teacher training pipeline is a promising direction for future work to generate stronger teacher policies.

References

[1] S. Brahmbhatt, A. Handa, J. Hays, and D. Fox, "Contactgrasp: Functional multi-finger grasp synthesis from contact," in *2019 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*. IEEE, 2019, pp. 2386–2393.

[2] T. Chen, J. Xu, and P. Agrawal, "A system for general in-hand object re-orientation," in *Conference on robot learning*. PMLR, 2022, pp. 297–307.

[3] O. M. Andrychowicz, B. Baker, M. Chociej, R. Jozefowicz, B. McGrew, J. Pachocki, A. Petron, M. Plappert, G. Powell, A. Ray *et al.*, "Learning dexterous in-hand manipulation," *The International Journal of Robotics Research*, vol. 39, no. 1, pp. 3–20, 2020.

[4] J. Luo, C. Xu, J. Wu, and S. Levine, "Precise and dexterous robotic manipulation via human-in-the-loop reinforcement learning," *Science Robotics*, vol. 10, no. 105, p. eads5033, 2025.

[5] A. Rajeswaran, V. Kumar, A. Gupta, G. Vezzani, J. Schulman, E. Todorov, and S. Levine, "Learning complex dexterous manipulation with deep reinforcement learning and demonstrations," *arXiv preprint arXiv:1709.10087*, 2017.

[6] I. Akkaya, M. Andrychowicz, M. Chociej, M. Litwin, B. McGrew, A. Petron, A. Paino, M. Plappert, G. Powell, R. Ribas *et al.*, "Solving rubik's cube with a robot hand," *arXiv preprint arXiv:1910.07113*, 2019.

[7] A. Petrenko, A. Allshire, G. State, A. Handa, and V. Makoviychuk, "Dexpbt: Scaling up dexterous manipulation for hand-arm systems with population based training," *arXiv preprint arXiv:2305.12127*, 2023.

[8] P. Liu, K. Zhang, D. Tateo, S. Jauhri, Z. Hu, J. Peters, and G. Chalvatzaki, "Safe reinforcement learning of dynamic high-dimensional robotic tasks: navigation, manipulation, interaction," in *2023 IEEE International Conference on Robotics and Automation (ICRA)*. IEEE, 2023, pp. 9449–9456.

[9] C. Yu and P. Wang, "Dexterous manipulation for multi-fingered robotic hands with reinforcement learning: A review," *Frontiers in Neurorobotics*, vol. 16, p. 861825, 2022.

[10] S. Ross, G. Gordon, and D. Bagnell, "A reduction of imitation learning and structured prediction to no-regret online learning," in *Proceedings of the fourteenth international conference on artificial intelligence and statistics*. JMLR Workshop and Conference Proceedings, 2011, pp. 627–635.

[11] R. Singh, A. Allshire, A. Handa, N. Ratliff, and K. Van Wyk, "Dextrah-rgb: Visuomotor policies to grasp anything with dexterous hands," *arXiv preprint arXiv:2412.01791*, 2024.

[12] J. Zhang and K. Cho, "Query-efficient imitation learning for end-to-end autonomous driving," *arXiv preprint arXiv:1605.06450*, 2016.

[13] W. Wang, F. Wei, L. Zhou, X. Chen, L. Luo, X. Yi, Y. Zhang, Y. Liang, C. Xu, Y. Lu *et al.*, "Unigrasptransformer: Simplified policy distillation for scalable dexterous robotic grasping," in *Proceedings of the Computer Vision and Pattern Recognition Conference*, 2025, pp. 12 199–12 208.

[14] L. Zhang, K. Bai, G. Huang, Z. Bing, Z. Chen, A. Knoll, and J. Zhang, "Contactdexnet: Multi-fingered robotic hand grasping in cluttered environments through hand-object contact semantic mapping," in *2025 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*. IEEE, 2025, pp. 8495–8502.

[15] L. Zhang, D. Zheng, K. Bai, Z. Bing, Z.-C. Marton, Z. Chen, A. C. Knoll, and J. Zhang, "Omnidexvlg: Learning dexterous grasp generation from vision language model-guided grasp semantics, taxonomy and functional affordance," *arXiv preprint arXiv:2512.03874*, 2025.

[16] E. Welte and R. Rayyes, "Interactive imitation learning for dexterous robotic manipulation: challenges and perspectives—a survey," *Frontiers in Robotics and AI*, vol. 12, p. 1682437, 2025.

[17] K. Menda, K. Driggs-Campbell, and M. J. Kochenderfer, "Ensembledagger: A bayesian approach to safe imitation learning," in *2019 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*. IEEE, 2019, pp. 5041–5048.

[18] M. Kelly, C. Sidrane, K. Driggs-Campbell, and M. J. Kochenderfer, "Hg-dagger: Interactive imitation learning with human experts," in *2019 International Conference on Robotics and Automation (ICRA)*. IEEE, 2019, pp. 8077–8083.

[19] H. Cai, Z. Peng, and B. Zhou, "Robot-gated interactive imitation learning with adaptive intervention mechanism," in *Proceedings of the 42nd International Conference on Machine Learning*, ser. Proceedings of Machine Learning Research, vol. 267. PMLR, 2025, pp. 6243–6256.

[20] R. Firoozi, J. Tucker, S. Tian, A. Majumdar, J. Sun, W. Liu, Y. Zhu, S. Song, A. Kapoor, K. Hausman, B. Ichter, D. Driess, J. Wu, C. Lu, and M. Schwager, "Foundation models in robotics: Applications, challenges, and the future," *The International Journal of Robotics Research*, vol. 44, no. 5, pp. 701–739, 2025.

[21] X. Han, S. Chen, Z. Fu, Z. Feng, L. Fan, D. An, C. Wang, L. Guo, W. Meng, X. Zhang, R. Xu, and S. Xu, "Multimodal fusion and vision-language models: A survey for robot vision," *Information Fusion*, vol. 126, p. 103652, 2026.

[22] J. Duan, W. Yuan, W. Pumacay, Y. R. Wang, K. Ehsani, D. Fox, and R. Krishna, "Manipulate-anything: Automating real-world robots using vision-language models," in *Proceedings of the 8th Conference on Robot Learning*, 2025, pp. 5326–5350.

[23] Y. Dai, J. Lee, N. Fazeli, and J. Chai, "RACER: Rich language-guided failure recovery policies for imitation learning," in *2025 IEEE International Conference on Robotics and Automation (ICRA)*. IEEE, 2025, pp. 15 657–15 664.

[24] Y. Wu, R. Tian, G. Swamy, and A. Bajcsy, "From foresight to forethought: VLM-in-the-loop policy steering via latent alignment," *arXiv preprint arXiv:2502.01828*, 2025.

[25] W. Wan, H. Geng, Y. Liu, Z. Shan, Y. Yang, L. Yi, and H. Wang, "Unidexgrasp++: Improving dexterous grasping policy learning via geometry-aware curriculum and iterative generalist-specialist learning," in *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2023, pp. 3891–3902.

[26] R. Hoque, A. Balakrishna, E. Novoseller, A. Wilcox, D. S. Brown, and K. Goldberg, "Thriftydagger: Budget-aware novelty and risk gating for interactive imitation learning," *arXiv preprint arXiv:2109.08273*, 2021.